

Software Requirement Specification (SRS)

Course Information

Course Title

Software Development III (CSE 4104)

Project Title

AI Study Material Generator & Assistant

Team Name

CSE4104-7C-T03

Section:

7C

Submitted By

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1. Introduction

1.1 Project Overview

AI-Powered Study Material Generator is a web-based educational platform that utilizes Artificial Intelligence to automatically generate study materials from uploaded learning resources such as PDFs, lecture notes, and text documents.

The system helps students save time by generating generate summaries, MCQs, flashcards, quizzes, and important questions. An AI-powered chatbot will also assist users in understanding topics and answering academic questions.

1.2 Problem Statement

Students often spend significant time preparing study materials manually from lecture notes and textbooks. Existing systems mainly provide static content and lack intelligent automation.

There is a need for a smart educational platform that can automatically generate personalized study resources and improve learning efficiency.

1.3 Objectives

- Develop an AI-powered study material generation platform.
- Generate summaries from educational content.
- Create MCQs and quizzes automatically.
- Generate flashcards for revision.
- Provide AI-assisted learning support.
- Improve student productivity and learning outcomes.

1.4 Scope

The system will serve students, teachers, and self-learners by providing AI-generated educational resources.

The scope includes:

- | | | |
|-----------------------|-----------------------|--------------------------|
| ➤ User authentication | AI content generation | PDF export functionality |
| ➤ Document upload | Quiz generation | AI chatbot support |

2. Functional Requirements

FR-01: User Registration

Users can create accounts using email and password.

FR-02: User Login

Registered users can securely log into the system.

FR-03: Profile Management

Users can view and update their profile information.

FR-04: Upload Study Materials

Users can upload PDF, DOCX, and TXT files.

FR-05: Document Storage

The system stores uploaded documents securely.

FR-06: Generate Summaries

The system generates concise summaries from uploaded documents.

FR-07: Generate MCQs

The system creates multiple-choice questions automatically.

FR-08: Generate Flashcards

The system generates flashcards for quick revision.

FR-09: Generate Quiz Sets

Users can generate quizzes based on uploaded content.

FR-10: AI Chatbot Support

Users can interact with an AI chatbot for study assistance.

FR-11: Download Materials

Users can download generated summaries, quizzes, and flashcards.

FR-12: Admin User Management

Admins can manage user accounts.

FR-13: Admin Document Monitoring

Admins can monitor uploaded documents and system activities.

3. Non-Functional Requirements

NFR-01: Performance

The system should generate results within a reasonable response time.

NFR-02: Security

User passwords must be encrypted before storage.

NFR-03: Availability

The system should be available 24/7 with minimal downtime.

NFR-04: Scalability

The platform should support increasing numbers of users and documents.

NFR-05: Reliability

The system should provide accurate and consistent outputs.

NFR-06: Usability

The interface should be easy to use and intuitive.

NFR-07: Responsiveness

The application should work on desktop, tablet, and mobile devices.

NFR-08: Maintainability

The system should be modular and easy to maintain.

4. User Roles

Admin

Responsibilities:

- Manage users
- Monitor uploaded documents
- Monitor AI activities
- Manage system operations

Student/User

Responsibilities:

- Register and login
 - Upload study materials
 - Generate summaries
 - Generate MCQs
 - Generate flashcards
 - Use AI chatbot
 - Download generated materials
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5. Use Cases

Use Case ID	Actor	Action	System Response
UC-01	Student	Registers with email & password	Creates account and sends welcome email
UC-02	Student	Logs in with valid credentials	Grants access to dashboard
UC-03	Student	Uploads a PDF file	Extracts text and prompts user to select generation type
UC-04	Student	Requests a summary	Calls AI API and displays a bullet-point summary
UC-05	Student	Requests MCQs	Generates 5-10 MCQs with answer key
UC-06	Student	Requests flashcards	Generates 5-10 flashcards
UC-07	Student	Asks a question to chatbot	Responds based on uploaded document content
UC-08	Student	Saves generated content	Stores it in MongoDB under user's profile
UC-09	Student	Downloads a quiz	Exports as a formatted PDF

UC-10	Admin	Logs in	Redirects to admin dashboard
UC-11	Admin	Views user activity	Displays analytics and logs

6. Initial Diagrams Required

Use Case Diagram

Actors:

- Admin
- Student/User

Main Use Cases:

- Register
- Login
- Upload Document
- Generate Summary
- Generate Notes
- Generate Quiz
- Use AI Chatbot
- Download Materials

7. ER Diagram

Entities

Users

- user_id (PK)
- name
- email
- password
- role

Documents

- document_id (PK)
- user_id (FK)
- file_name
- file_type
- upload_date

Generated_Materials

- material_id (PK)
- document_id (FK)
- material_type
- generated_content

Quiz

- quiz_id (PK)
- document_id (FK)
- questions

Chat_History

- chat_id (PK)
- user_id (FK)
- message
- response

8. Technology Stack

Frontend

- React.js
- HTML
- CSS
- JavaScript

Backend

- Node.js
- Express.js

Database

- MongoDB

Authentication

- JWT

AI Integration

- OpenAI API

Deployment

- Vercel
- Render

9.Expected Outcome

The AI Study Material Generator will help students save time by automatically creating study materials from educational documents. The platform will improve learning efficiency, provide personalized study assistance, and enhance academic productivity.

10. Conclusion

The AI-Powered Study Material Generator aims to simplify academic learning by automating the creation of summaries, notes, quizzes, and personalized recommendations using Artificial Intelligence. The system will improve learning efficiency, reduce study preparation time, and provide an intelligent educational support platform for students.